



Display Unit

User Manual

PCS corporation

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Republic of Korea

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1. RTU Main Specifications

1.1 Manufacturer : PCS corporation

1.2 Model Name : AMI-RTU

1.3 Product Specifications and Standards

1) CPU : 180 MHz, 32 bit

2) RAM : Flash Memory 512 M

3) Storage : Micro-SD

4) Ethernet : TCP/IP Support

5) RS232C(RS485) : 8CH (RS485:Converters should be used)

6) Analog Input (4~20 mA) : 8CH

7) Analog Output (4~20 mA) : 2CH

8) Digital Input : 8CH

9) Digital Output : 7CH

10) USB : 1CH (Save data)

11) Screen : 8" Full Touch Screen (Power Saving Mode)

12) Material : ABS, Metal

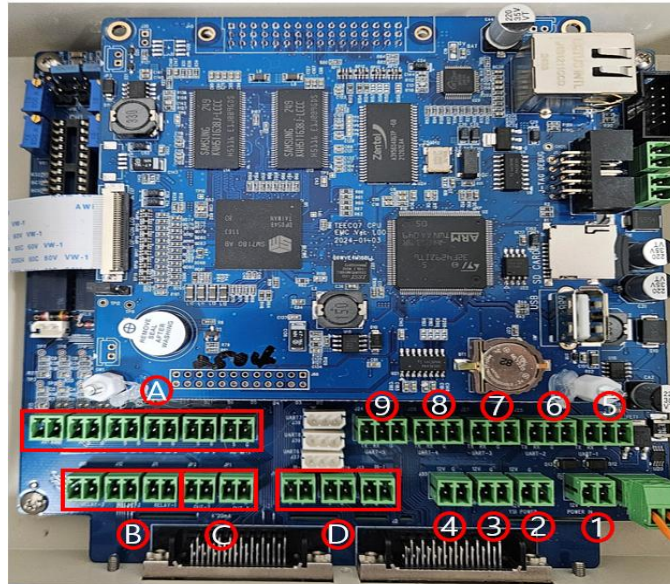
13) Mounting Method : Panel(wall) Installation

14) Power Supply : DC 12V, 700mA

15) Dimensions : 250(W) x 80(D) x 205(H) mm

16) Power Consumption : 1.8 W (Low Power), 4 W (MAX)

1.4 Input/Output and Wiring



1) 12V Power In

+	-
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2) 12V Power out for COD 12 V Wiring

3) 12V Power out for TSS, pH

+	-
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4) 12V Power out for NHN

5) RS485 for COD RS485 Wiring

6) RS485 for TSS

7) RS485 for pH

8) RS485 for NHN

9) Modbus (RS485 output)

A) Analog In (4 - 20 mA) / Not Used

B) Digital Out (Relay) / Not Used

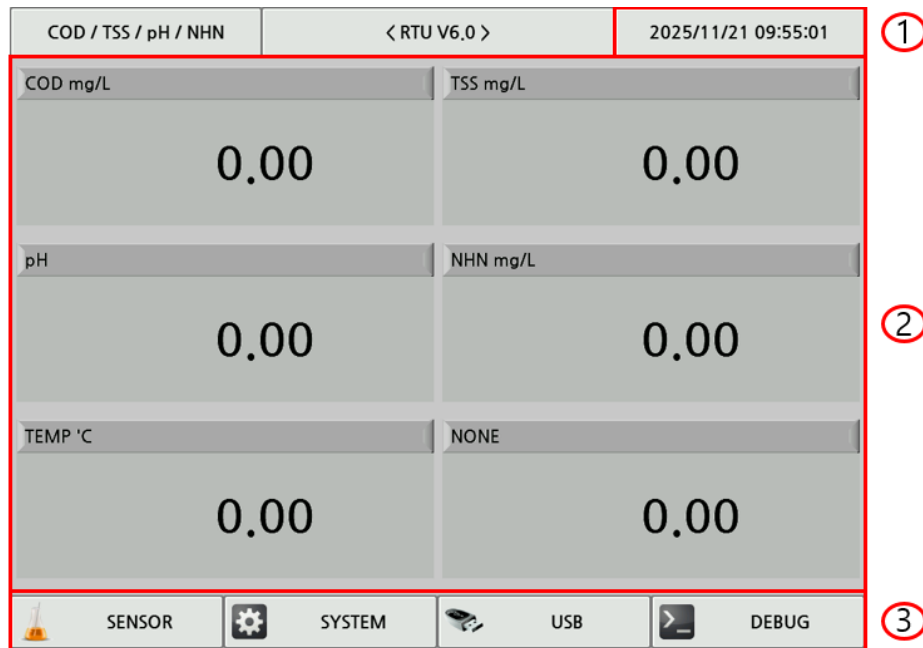
C) Analog Out (4 - 20 mA) / Not Used

D) Digital In / Not Used

RS485	RS485	
A	B	

2. RTU Settings

2.1 Main Screen



1) RTU Current Time Display

2) Real-time data : The data measured by the sensor is received at second intervals.

3) Menu Bar

① Sensor : Communication settings for sensors (RS232 / RS485)

② SYSTEM : RTU Settings and Sensor Calibration

③ USB : Data is saved on USB when USB is connected to the port and the button is touched.

④ DEBUG : Check the status of data storage and communication (Used for inspection)

3) RTU Current Time Display

2.2 SENSOR Settings

SENSOR			< RTU V6.0 >		2025/11/19 13:29:10	
USART1			ANALOG INPUT		4~20mA OUTPUT	
9600	RS485	COD	0.0	CAL START	CH1	CH2
USART2			CH1	CH2	CH3	CH4
9600	RS485	TSS	CH5	CH6	CH7	CH8
USART3			CH9	CH10	NONE	
9600	RS485	pH	item	NONE	CAL START	
USART4			min mV	357.0	Name	NONE
9600	RS485	NHN	max mV	1792.0	Zero	0.0
External communication			min PV	0.0	Span	0.0
LAN	DISABLE		max PV	100.0	4~20mA	CALOUT
					③	
①			②		HOME	
					④	

- ① Sensor Communication Settings
 - 1) USART1 : 9600 / RS485 / COD
 - 2) USART2 : 9600 / RS485 / TSS
 - 3) USART3 : 9600 / RS485 / pH
 - 4) USART4 : 9600 / RS485 / NHN
 - 5) External communication : LAN / DISABLE
- ② ANALOG INPUT (4 – 20 mA) : Not Used
- ③ ANALOG OUTPUT (4 – 20 mA) : Not Used
- ④ Return to Main Screen

2.3 SYSTEM Settings

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20,104,219,1	time set	start	COD	start
s-sub	255,255,255,248	reboot(hour)	25	TSS	start
s-gate	20,104,219,3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0,128,220,0,171,16	factory reset		MODBUSID	1
c-ip	0,0,0,0	reset	start	3	
c-sub	0,0,0,0	file backup	start		
c-gate	0,0,0,0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0,128,220,0,171,16	color	6		
				HOME	

① ② ④

① LAN Settings : Enter S-IP / S-SUB / S-GATE / S-PORT

- Not Used

② Hardware Settings

1) Time : Set RTU Time

2) Reboot (Hour) : Set RTU Automatic Reboot timing

- Set to 25 when not in use

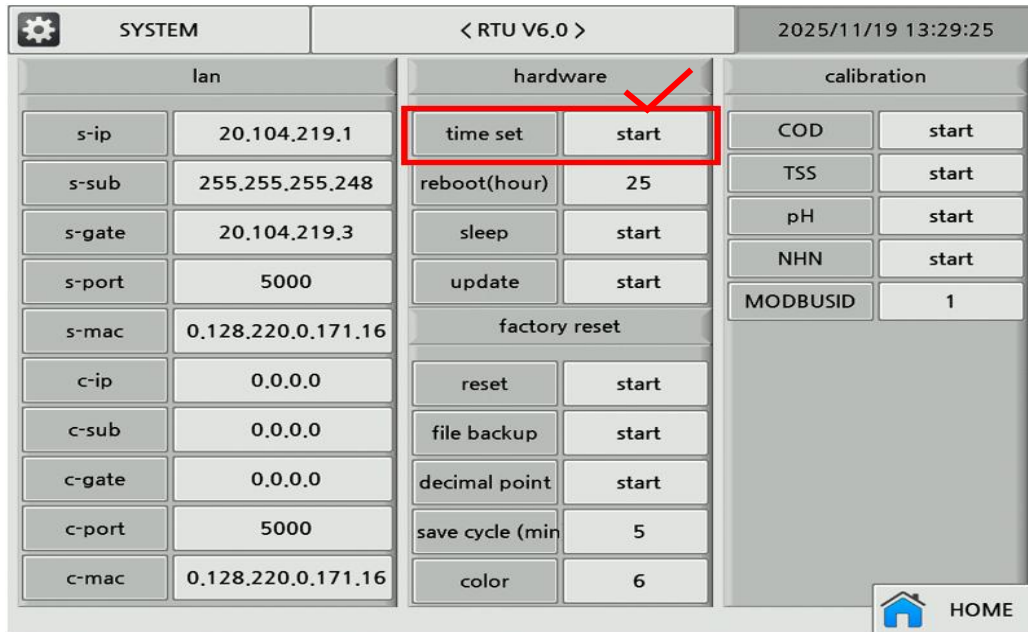
3) Sleep : Enter Power Saving Mode

4) Update : Not Used

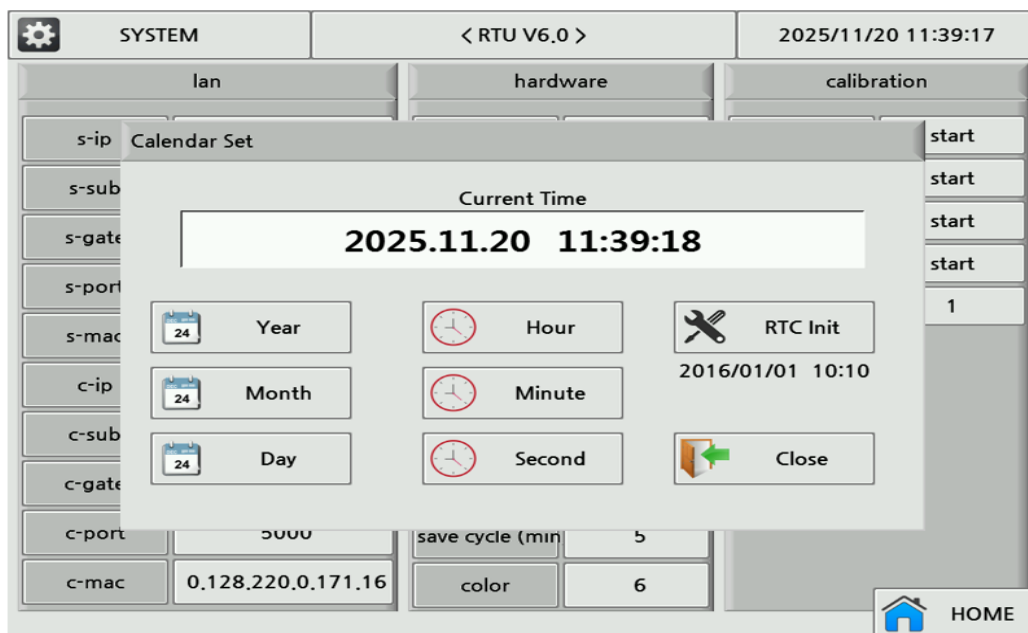
5) Factory Reset : Reset data stored in RTU SD card

- 6) Create Backup File : Not Used
- 7) Decimal Point : Can be set from 0 to 4 decimal places
- 8) Save Cycle (min) : Can be set to 1, 5, or 10 minute intervals
- 9) COLOR : Change RTU background color
- ③ Calibration : Calibrate each sensor
- ④ Return to Main Screen

2.3.1) Time Settings



① Click 'start'



② After setting the time, touch Close.

2.3.2) Automatic Reboot (Hour) Settings

The screenshot shows the RTU V6.0 SYSTEM menu. The 'hardware' section is active, and the 'reboot(hour)' setting is highlighted with a red box. A red checkmark is placed over the 'start' button next to it. The 'lan' section shows IP addresses for serial and console ports. The 'calibration' section shows various sensors and their status.

lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

① Click '25'

The screenshot shows the RTU V6.0 SYSTEM menu with the 'reboot(hour)' setting. A numeric keypad overlay is displayed, showing the number '25' entered. The keypad has buttons for digits 0-9, a decimal point, a plus/minus sign, and an 'ENT' button. The 'lan' and 'calibration' sections are visible in the background.

lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.2	reboot(hour)	25	TSS	start
s-gate	20.104.219.3			pH	start
s-port	5000			NHN	start
s-mac	0.128.220.0.171.16			MODBUSID	1
c-ip	0.0.0.0				
c-sub	0.0.0.0				
c-gate	0.0.0.0				
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

① Set the desired RTU Automatic Reboot time, and touch ENT.

- If not used, enter 25
- Example: Enter 12 to automatically reboot every 12 hours

2.3.3) Power Saving Mode Settings

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25 ✓	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click 'start'

SYSTEM		< RTU V6.0 >		2025/11/19 13:35:46	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

sleep

Power saving execution.
Run in 60 seconds

OK

② Click OK and the screen will turn off in 60 seconds

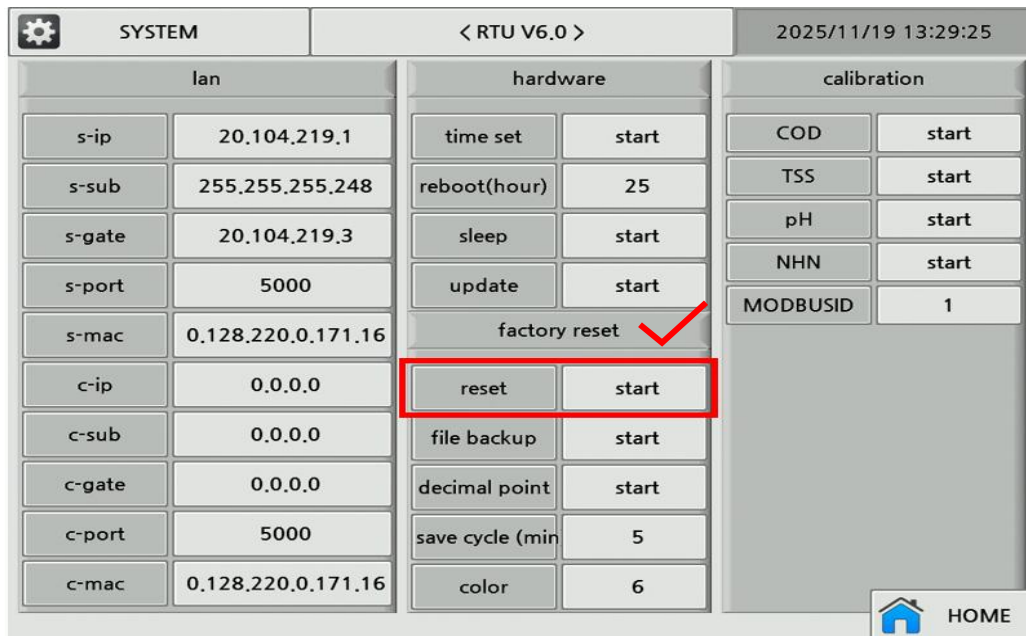
2.3.4) Update

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

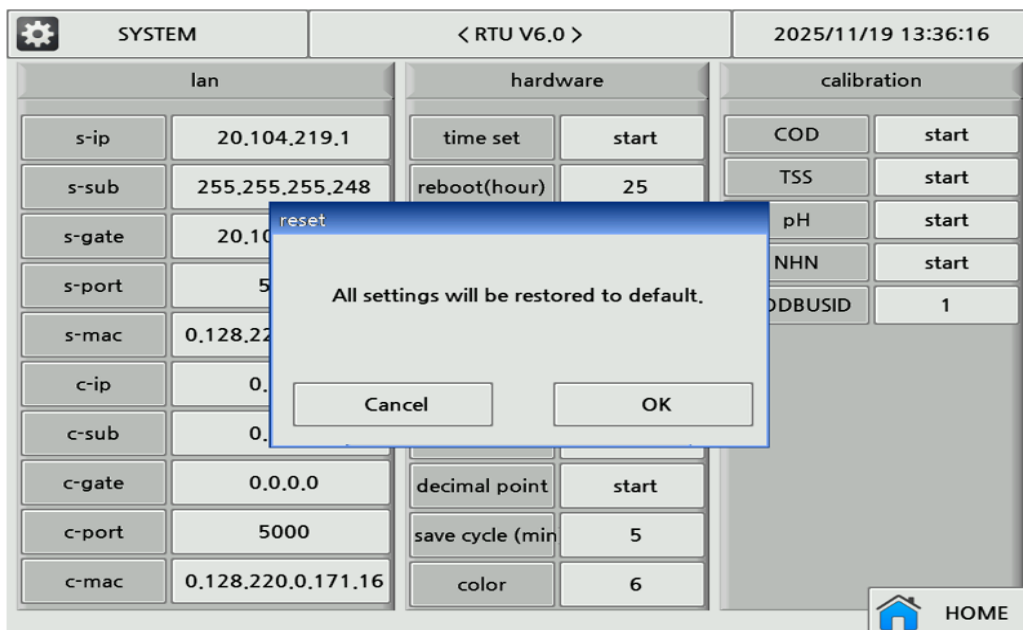
 HOME

① Update : Not Used

2.3.5) Factory Reset



① Click 'start'



② After touching 'Start' on the factory reset menu, the 'All settings will be restored to default.' message will appear. When the message is displayed, Touch OK.

2.3.6) Create Backup File

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

 HOME

① Create Backup File : Not Used

2.3.7) Decimal Places Settings

SYSTEM < RTU V6.0 > 2025/11/19 13:29:25

lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click 'start'

Decimal digit setting

item	point	item	point
COD mg/L	2	NONE	NONE
TSS mg/L	2	NONE	NONE
pH	2	NONE	NONE
NHN mg/L	2	NONE	NONE
TEMP 'C	2	NONE	NONE
NONE	2	NONE	NONE
NONE	2	NONE	NONE
NONE	2	NONE	NONE
NONE	2	NONE	NONE
NONE	2	NONE	NONE
NONE	2	NONE	NONE
NONE	2	NONE	NONE

HOME

② Touch to set the decimal places of the item.

2.3.8) Save Interval (minutes) Settings

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

- ① Click '5'
- ② Intervals of 1, 5, 10 minutes can be set

2.3.9) RTU Background Color Settings

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click '6'

② Background color can be changed from 1 to 6

3. Calibration

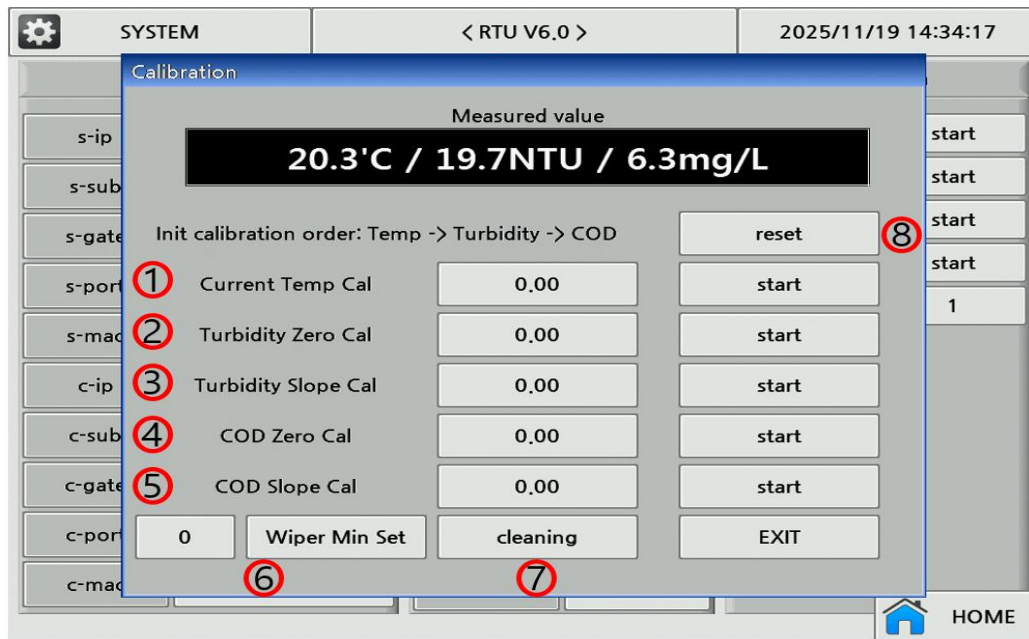
3.1 COD

- COD Item factory reset, calibration, and cleaning

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration ✓	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click 'start'



② Screen information

1. Temperature Calibration
2. Turbidity zero Calibration
3. Turbidity span (slope) Calibration
4. COD zero Calibration
5. COD span(slope) Calibration
6. Wiper Operation Interval (minutes)
 - Configurable from 6 minutes to 6,000 minutes
 - Enter a number and click 'Wiper Min Set'
7. Wiper Operation
8. Factory reset

③ Calibration Information

***Temperature > Turbidity > COD must always be the sequence followed**

Order	Item	Solution
1	Temperature	Current temperature (using thermometer, etc.)
2	Turbidity zero	0 NTU
3	Turbidity span(slope)	800 NTU
4	COD zero	0 mg/L COD solution
5	COD span(slope)	1,000 mg/L COD solution

1. Temperature Calibration

- 1-1. Immerse the sensor in distilled water.
- 1-2. Enter the current temperature in the '0.00' field next to 'Current Temp Cal'
- 1-3. Once the value is stable, click 'start', it will change to 'stop'
- 1-4. After 1 minute, click 'stop'
- 1-5. First, confirm the temperature in the black box matches the entered value. Then proceed to the next step

2. Turbidity zero Calibration

- 2-1. Immerse the sensor in 0 NTU solution.
- 2-2. Enter '0' in '0.00' next to 'Turbidity Zero Cal'.
- 2-3. Once the value is stable, click 'start', it will change to 'stop'
- 2-4. After 1 minute, click 'stop'
- 2-5. First, confirm the turbidity in the black box matches the entered value. Then proceed to the next step

3. Turbidity span(slope) Calibration

- 3-1. Immerse the sensor in 800 NTU solution.
- 3-2. Enter '800' in '0.00' next to 'Turbidity Zero Cal'.
- 3-3. Once the value is stable, click 'start', it will change to 'stop'
- 3-4. After 1 minute, click 'stop'
- 3-5. First, confirm the turbidity in the black box matches the entered value. Then proceed to the next step

4. COD zero adjustment

- 4-1. Immerse the sensor in 0 mg/L COD solution.
- 4-2. Enter '0' in the '0.00' field next to 'COD Zero Cal'
- 4-3. Once the value is stable, click 'start', it will change to 'stop'
- 4-4. After 1 minute, click 'stop'
- 4-5. First, confirm the COD in the black box matches the entered value. Then proceed to the next step

5. COD span(slope) Calibration

- 5-1. Immerse the sensor in 1,000 mg/L COD solution
- 5-2. Enter '0' in the '0.00' field next to 'COD Slope Cal'
- 5-3. Once the value is stable, click 'start', it will change to 'stop'
- 5-4. After 1 minute, click 'stop'
- 5-5. First, confirm the COD in the black box matches the entered value. Then click the 'EXIT' button

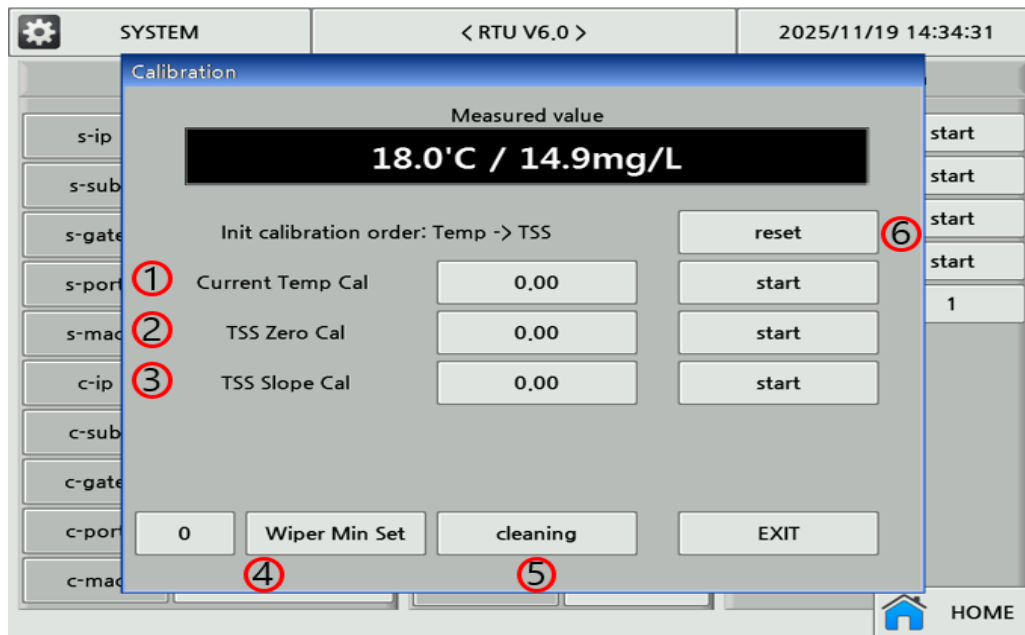
3.2 TSS

- TSS Item factory reset, calibration, and cleaning

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click 'start'



②Screen Information

1. Temperature Calibration
2. TSS zero Calibration
3. TSS span(slope) Calibration
4. Wiper Operation Interval (minutes)
 - Configurable from 6 minutes to 6,000 minutes
 - Enter a number and click 'Wiper Min Set'
5. Wiper Operation
6. Factory reset

③ Calibration Information

***Temperature > TSS must always be the sequence followed**

Order	Item	Solution
1	Temperature	Current temperature (using thermometer, etc.)
2	TSS zero	0 mg/L
3	TSS span(slope)	200 mg/L

1. Temperature Calibration

- 1-1. Immerse the sensor in distilled water.
- 1-2. Enter the current temperature in the '0.00' field next to 'Current Temp Cal'
- 1-3. Once the value is stable, click 'start', it will change to 'stop'
- 1-4. After 1 minute, click 'stop'
- 1-5. First, confirm the temperature in the black box matches the entered value. Then proceed to the next step

2. TSS zero adjustment

- 2-1. Immerse the sensor in 0 mg/L TSS solution.
- 2-2. Enter '0' in the '0.00' field next to 'TSS Zero Cal'
- 2-3. Once the value is stable, click 'start', it will change to 'stop'
- 2-4. After 1 minute, click 'stop'
- 2-5. First, confirm the TSS in the black box matches the entered value. Then proceed to the next step

3. TSS span(slope) Calibration

3-1. Immerse the sensor in 200 mg/L solution.

3-2. Enter '200' in the '0.00' field next to 'TSS Slope Cal'

3-3. Once the value is stable, click 'start', it will change to 'stop'

3-4. After 1 minute, click 'stop'

3-5. First, confirm the TSS in the black box matches the entered value. Then click the 'EXIT' button

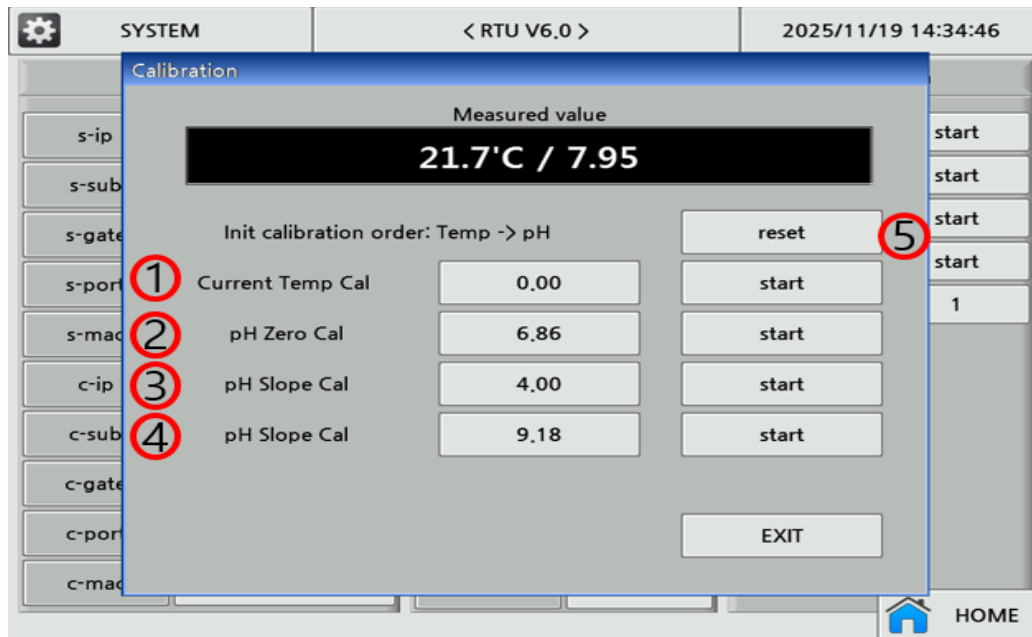
3.3 pH

- Factory reset and calibration of the pH parameter

SYSTEM		< RTU V6.0 >		2025/11/19 13:29:25	
lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click 'start'



②Screen Information

1. Temperature Calibration
2. pH zero Calibration
3. pH span(slope) Calibration
4. pH span(slope) Calibration
5. Factory reset

③ Calibration Information

***Temperature > pH 6.86 > pH 4.00 > pH 9.18 must always be the sequence followed**

Order	Item	Solution
1	Temperature	Current temperature (using thermometer, etc.)
2	pH zero	6.86
3	pH span(slope)	4.00
4	pH span(slope)	9.18

1. Temperature Calibration

- 1-1. Immerse the sensor in distilled water.
- 1-2. Enter the current temperature in the '0.00' field next to 'Current Temp Cal'
- 1-3. Once the value is stable, click 'start', it will change to 'stop'
- 1-4. After 1 minute, click 'stop'
- 1-5. First, confirm the temperature in the black box matches the entered value. Then proceed to the next step

2. pH zero Calibration

- 2-1. Immerse the sensor in pH 6.86 solution.
- 2-2. Once the value is stable, click 'start', it will change to 'stop'
- 2-3. After 1 minute, click 'stop'
- 2-4. First, confirm the pH value in the black box is set to '6.86'. Then proceed to the next step

3. pH slope Calibration

3-1. Immerse the sensor in pH 4.00 solution.

3-2. Once the value is stable, click 'start', it will change to 'stop'

3-3. After 1 minute, click 'stop'

3-4. First, confirm the pH value in the black box is set to '4.00'.

Then proceed to the next step

4. pH slope Calibration

4-1. Immerse the sensor in pH 9.18 solution.

4-2. Once the value is stable, click 'start', it will change to 'stop'

4-3. After 1 minute, click 'stop'

4-4. First, confirm the pH in the black box is set to '9.18'.

Then click the 'EXIT' button

3.4 NHN

- Factory reset and calibration of the NHN parameter

SYSTEM

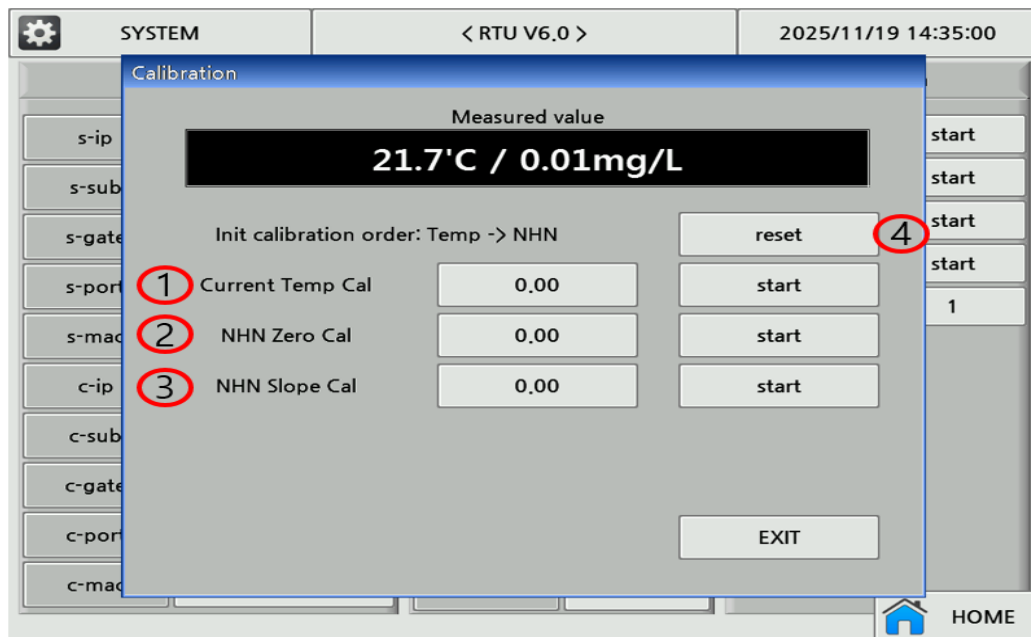
< RTU V6.0 >

2025/11/19 13:29:25

lan		hardware		calibration	
s-ip	20.104.219.1	time set	start	COD	start
s-sub	255.255.255.248	reboot(hour)	25	TSS	start
s-gate	20.104.219.3	sleep	start	pH	start
s-port	5000	update	start	NHN	start
s-mac	0.128.220.0.171.16	factory reset		MODBUSID	1
c-ip	0.0.0.0	reset	start		
c-sub	0.0.0.0	file backup	start		
c-gate	0.0.0.0	decimal point	start		
c-port	5000	save cycle (min)	5		
c-mac	0.128.220.0.171.16	color	6		

HOME

① Click 'start'



②Screen Information

1. Temperature Calibration
2. NHN zero Calibration
3. NHN span(slope) Calibration
4. Factory reset

③ Calibration Information

***Temperature > NHN must always be the sequence followed**

Order	Item	Solution
1	Temperature	Current temperature (using thermometer, etc.)
2	NHN zero	1 mg/L NHN solution
3	NHN span(slope)	10 mg/L NHN solution

1. Temperature Calibration

- 1-1. Immerse the sensor in distilled water.
- 1-2. Enter the current temperature in the '0.00' field next to 'Current Temp Cal'
- 1-3. Once the value is stable, click 'start', it will change to 'stop'
- 1-4. After 1 minute, click 'stop'
- 1-5. First, confirm the temperature in the black box matches the entered value. Then proceed to the next step

2. NHN zero adjustment

- 2-1. Immerse the sensor in 1 mg/L NHN solution.
- 2-2. Once the value is stable, click 'start', it will change to 'stop'
- 2-3. After 1 minute, click 'stop'
- 2-4. First, confirm the NHN value in the black box is set to '1 mg/L'. Then proceed to the next step

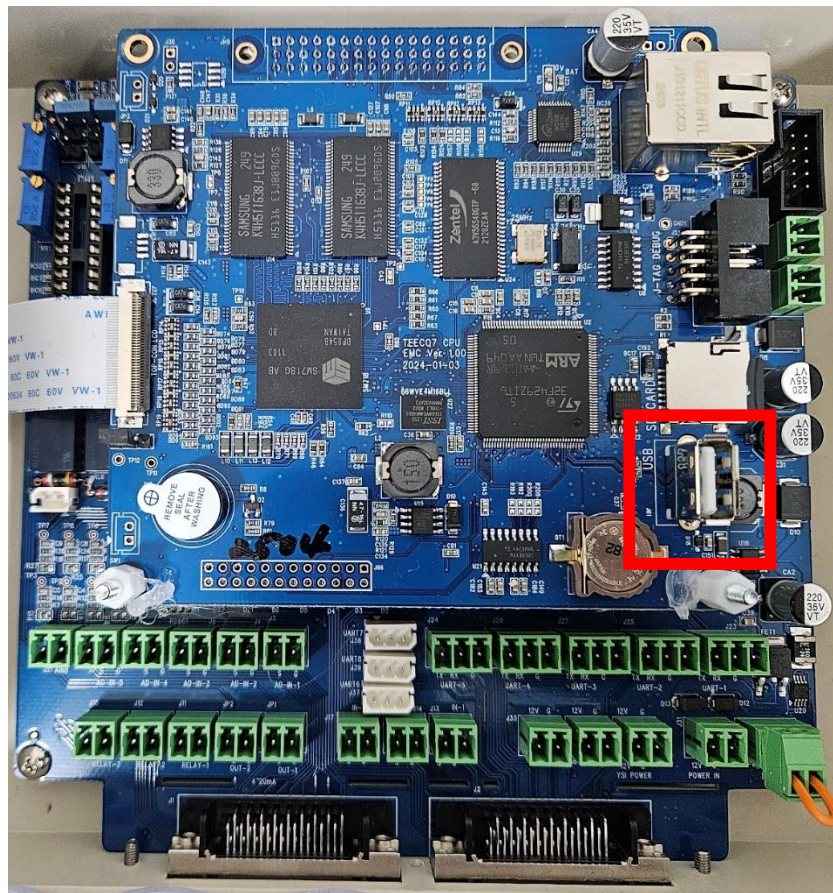
3. NHN slope Calibration

- 3-1. Immerse the sensor in 10 mg/L NHN solution.
- 3-2. Once the value is stable, click 'start', it will change to 'stop'
- 3-3. After 1 minute, click 'stop'
- 3-4. First, confirm the NHN value in the black box is set to '10 mg/L'. Then click the 'EXIT' button

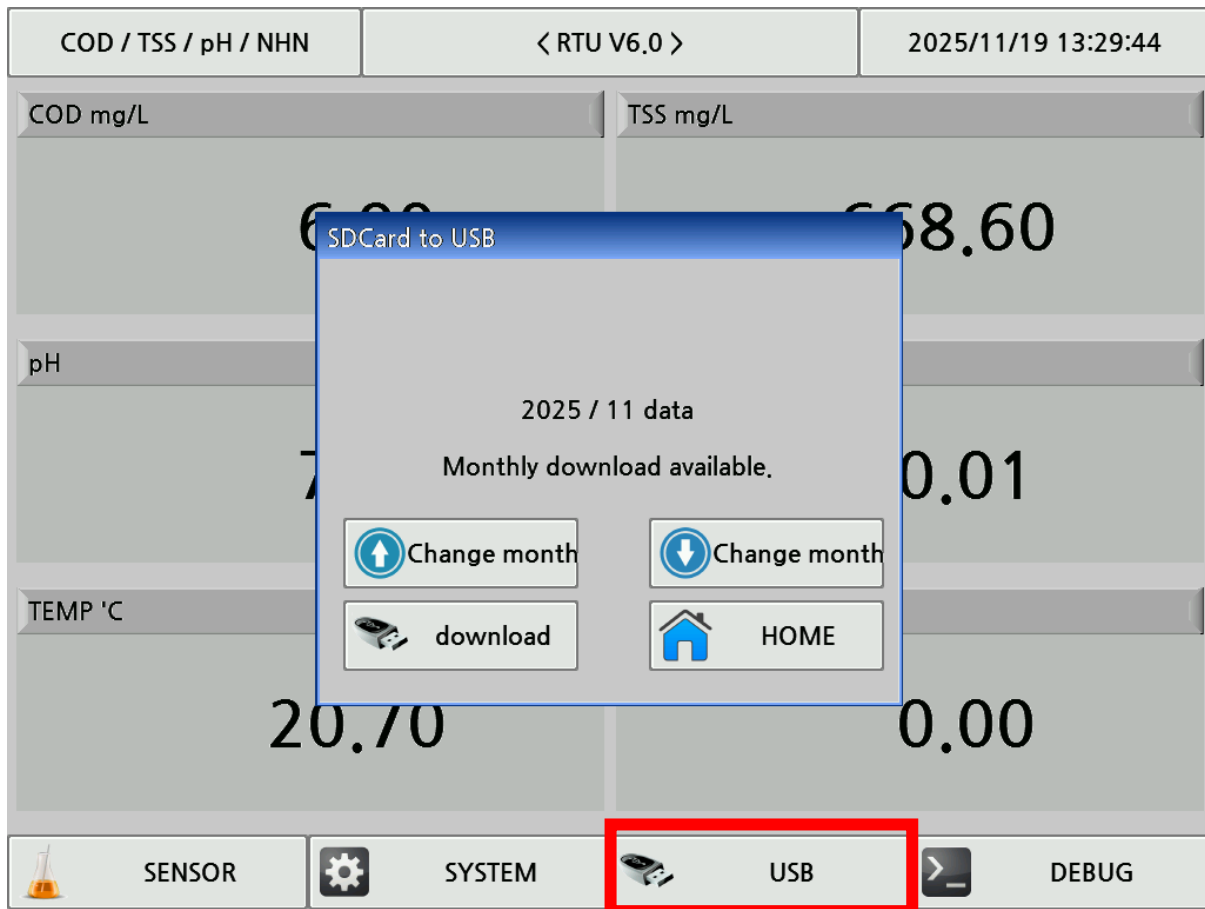
4. Download Data

The collected data can be downloaded via USB.

4-1. Download Data



1) Insert the USB into the USB port.



- 2) On the main screen, click 'USB'.
- 3) On this screen, select the month of the data to download.
- 4) If the USB is recognized, 'press the [download] button to start the download.' text is displayed. (Use USB 2.0)
- 5) Press download.